DO NOW — RECALL

Lesson 115 · The Stages of Compilation

Name: _____ Class: _____ Date: _____

1. What tool converts assembly mnemonics into machine instructions?

2. Why must a binary search operate on ordered data?

3. An OR gate has inputs 0 and 1. What is the output?

4. What is the effect of enqueue(X), enqueue(Y), dequeue() on an initially empty FIFO queue?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

An assembler.

2. Model answer

It discards half the search space using an ordering comparison.

3. Model answer

1.

4. Model answer

X is removed; Y remains at the front.

By the end, you can…

• Characters become tokens such as identifiers, numbers and operators.

• A parser checks grammar and builds structural representation.

• Target code is produced; valid transformations improve performance while preserving meaning.

Vocabulary — know these terms

lexer — Recognises tokens from character stream.

parser — Checks tokens against grammar.

code generation — Production of target instructions.

optimisation — Meaning-preserving transformation to improve an objective.

NEW LEARNING

Read, annotate and explain

Lexical analysis

Characters become tokens such as identifiers, numbers and operators.

Syntax analysis

A parser checks grammar and builds structural representation.

Code generation and optimisation

Target code is produced; valid transformations improve performance while preserving meaning.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Explain the stages for the expression 'total = cost + 2' in a high-level source file.

Model answer

Lexer identifies identifier, assignment, identifier, plus, integer. Parser checks grammatical structure. The compiler creates suitable target instructions; optimisation might fold constants only where semantics permit.

Check: link each claim to the scenario and use precise OCR terminology.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Describe lexical analysis, syntax analysis and code generation. [6]

Question 2

For 'score = score + 1', name five token categories or concrete tokens. [5]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate whether an optimiser may replace 'x = a + 0' with 'x = a', considering language semantics. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A syntax analyser is responsible primarily for reading characters into tokens.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.